PeriGen
Advanced Perinatal Systems

PeriWatch Surveillance™ / PeriWatch Surveillance in Vigilance Mode

2025.1.000

System Requirements

Table of Contents

System Definitions 3
Recommended Standard / Enterprise Model (Multi-Facilities) Server Requirements 3
Application Server (All-in-one application services) VM Specifications 3
Dedicated SQL Server VM Specifications (if applicable) 4
Dedicated HL7 Server (Monitored locations > 300 beds or iBus interface) VM Specifications 4
PeriWatch All-In-One - Test/Training 5
Certificate Requirements 7
PeriWatch Surveillance – Small Site Configuration VM Specification 7
Virtual Servers 8
Archive Storage (Only applicable for full Surveillance servers) 8
Client / Workstation Requirements 8
Thin-Client Technologies 9
Network (For Workstations and Data Acquisition) 9
Tracing Data Acquisition 10
Fetal Monitor Compatibility 11
DAS Specifications 11
Anti-Virus Exclusions 15

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

System Definitions

The PeriWatch Surveillance system can be combined in 3 main configurations:

- Full Surveillance without Vigilance mode – Standard surveillance system with no Vigilance Early Warning system components
- Full Surveillance with integrated Vigilance components – Like the above with Vigilance Cues, Curve and HUB
- Standalone Vigilance – Core data acquisition and interfaces in "Vigilance Mode", meaning all Vigilance components complementing a pre-existing 3rd party surveillance system and EMR.

Recommended Standard / Enterprise Model (Multi-Facilities) Server Requirements

- A full enterprise implementation of PeriWatch Surveillance and Vigilance's Decision Support modules requires a base configuration of 3 server VMs as follows:
    - **Application VM:** PeriWatch Surveillance Application/Vigilance Decision Support/HL7 Interface VM (a dedicated HL7 integration server is required if shared with multiple App servers or when interfaced with Cerner iBus, refer LAB00005 and LAB00019)
    - **Database VM:** SQL Server DB Server or a clustered AG environment hosted by the customer
    - **Test/Train VM:** PeriWatch All-In-One (SQL & App) Test/Train/Custom Reports VM

- A configuration hosting PeriWatch Surveillance on one server and PeriWatch Vigilance on another server is also supported (DB on cluster or farm).
- See "Small Site configuration" section for sites <1000 births annually and <10 monitored locations.
- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined with the customer during implementation planning stage.
- Physical servers (if required) must follow custom/sizing configuration by PeriGen.
- See certificate requirement section for secure system configuration (default).

Application Server (All-in-one application services) VM Specifications

# of Monitored locations (1)	50	100	200	500
# of vCPUs	4	8	12	12
GB RAM	12	16	20	24
Volume C: OS	100 GB			

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Volume D: Applications	100 GB

Dedicated SQL Server VM Specifications (if applicable)

# of Monitored locations (1)	50	100	200	500
# of vCPUs	4	8	12	12
GB RAM	12	16	20	24
Volume C: OS	Customer defined default size			
Volume E: SQLData(2)	200 GB	300 GB	500 GB	750 GB
Volume F: SQLTransactionLog	50 GB			
Volume G: SQLBackup	According to customer maintenance plan			
Volume H: SQLTempDB	50 GB			

Table A: Bandwidth and performance requirements If SQL server is hosted on a separate cluster or farm:

# of Monitored locations (1)	50	100	200	500
Transactions / sec	10-50	20-100	40-200	60-300
Disk Reads (MB/sec)	2-60	4-120	8-240	12-360
Disk Writes (MB/sec)	1-20	2-40	5-80	8-120

(1) The CPU necessary on the Server comes from estimating the number of workstations simultaneously using the PeriWatch Surveillance Client. This number is often difficult to compute. Monitored locations are easier to determine thus the table above shows Monitored locations. However, it assumes that each Monitored location is running a PeriWatch Surveillance Client and there are no more than 10% additional workstations simultaneously doing the same. If it is anticipated that this will not be the case, then it will become necessary to adjust the CPU resources accordingly.

(2) SQL Data drive is the minimum requirement; more accurate requirement can be calculated based on number of births or visits per patient. Approximately 5MB is the total space required per visit.

Dedicated HL7 Server (Monitored locations > 300 beds or iBus interface) VM Specifications

# of vCPUs	8
GB RAM	16
Volume C: OS	60 GB

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Volume D: Data Store	200 GB

System Software Requirements

Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
Microsoft Windows Server 2016/2019/2022
Internet Information Services (IIS)
ASP.NET
Windows Authentication
MS SQL 2016 SP2 or 2019 (Standard edition or higher) or 2022, all with Reporting Services (clustered or farm setup supported according to PeriGen guidelines)
VC++ Redistributable: 2010 SP1, 2013, 2015
.Net 3.5 SP1, .Net 4.8
.NET 8.0.1 Windows Hosting (formerly .Net Core)

NOTE:

Customer upgrading from previous versions where the production application is running on 2 servers, this configuration is still supported providing the requirements for the Main Server are met for current servers or are equivalent. Please validate with your PeriGen technical representative.
For configuration exceeding the number of monitored areas, please contact your PeriGen technical representative for an updated, approved configuration.

PeriWatch All-In-One - Test/Training

A minimum of 1 TEST/Train server is mandatory. More servers can be allocated as needed
4 vCPUs
12 GB RAM
Storage requirements per volume:
C: Operating system - 60 GB
D: Applications/SQLData/Logs - 200 GB

Gigabit Ethernet
Microsoft Windows Server 2016/2019/2022
Internet Information Services (IIS) installed
ASP.NET
Windows Authentication
MS SQL 2016 SP2 or 2019 (Standard edition or higher) or 2022, all with Reporting Services

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

VC++ Redistributable: 2010 SP1, 2013, 2015
.Net 3.5 SP1, .Net 4.8
.NET 8.0.1 Windows Hosting (formerly .Net Core)

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Certificate Requirements

The customer must provide a server-side SSL certificate to encrypt all data in transit across the system.
The customer is responsible for all certificate renewal management.
Wildcard certificates are supported.
A certificate for SQL TDE is optional for full DB encryption on SQL Enterprise or Data Center versions.
The SSL certificate minimal requirements are:

Signature Algorithm:	SHA256RSA
Signature hash algorithm:	SHA256
Subject:	<FQDN of server> (ex: VigilanceApp01.Hospital.Org)
Subject Alternative Name:	<FQDN of server> (ex: VigilanceApp01.Hospital.Org)
Public Key:	RSA (2048 Bits)
Key Usage:	Digital Signature, Key Encipherment (a0)
Enhanced Key Usage:	Server Authentication (1.3.6.1.5.5.7.3.1) Client Authentication (1.3.6.1.5.5.7.3.2)

For SQL TDE, the certificate must have the AT_KEYEXCHANGE set to 1

Small Site Configuration (<1000 births annually and <10 monitoring locations)

- A small site configuration application to <1000 birth and under 10 monitored locations.
- With such a small load, all application modules can be consolidated on the primary Application/DB production server.
- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined at system implementation planning stage.
- If required, physical server(s) option should be equivalent to the VM specifications below.
- Any deviation from the Small Site definition above will require a custom sizing by PeriGen.
- A test server is still mandatory for any configuration.

PeriWatch Surveillance – Small Site Configuration VM Specification

# of Monitored locations	10
# of vCPUs	4
GB RAM	12
Volume C: OS	60 GB
Volume D: Applications	40 GB
Volume E: SQLData	200 GB

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Volume F: SQLTransactionLog	50 GB
Volume G : SQLBackup	Customer defined

System Software Requirements

Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
Microsoft Windows Server 2016/2019/2022
Internet Information Services (IIS) installed
ASP.NET
Windows Authentication
MS SQL 2016 SP2 or 2019(Standard edition or higher) or 2022, all with Reporting Services
VC++ Redistributable: 2010 SP1, 2013, 2015
.Net 3.5 SP1, .Net 4.8
.NET 8.0.1 Windows Hosting (formerly .Net Core)

Virtual Servers

- VMware (6.X and above) and Hyper-V is currently supported for all the configurations above as long as it is part of an enterprise virtualization infrastructure (not on standalone server).
- Other platforms will require extra validation services.

Archive Storage (Only applicable for full Surveillance servers)

SAN or NAS storage can be used for long term encrypted archive storage (customer supplied)
Archive storage requirements is based on storage requirement of 5 MB per birth.
Illustration:
8000 births annually for the entire enterprise
8000 births per year x 5MB = 40GB/year.

Client / Workstation Requirements

Microsoft Windows 10 Version 22H2 or Microsoft Windows 11 Version 22H2 and above
Minimum 2GB RAM, 4GB recommended. 250 MB of available memory should be reserved for PeriWatch Surveillance modules.
1280x1024 (4:3) is the minimal resolution supported by the system.
1024x768 (4:3) is supported with PeriWatch but not supported when Cues & Curve modules are purchased.
Higher resolutions in 16:9 or 16:10 aspect rations are supported only in native resolution (non-native resolutions or aspect ratios will result in a stretched representation of fetal tracings and are not supported)
.NET Framework 4.8
Microsoft Visual C++ Redistributable 2010 SP1 (x86/x64) and 2015 (x86/x64).

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

For Vigilance HUB Admin client: Microsoft Visual C++ Redistributable 2013
For Vigilance HUB: Supported web browser with JavaScript support enabled such as Google Chrome (recommended) or MS Edge (Chromium).

The following are only applicable for full Surveillance implementations:

Wireless network is not recommended for the PeriWatch Surveillance application. If required, PeriGen will collaborate with the customer to confirm that PeriWatch application performs as expected on the site's wireless environment.
Large Screen displays of non-interactive multi-tracing Surveillance or Census screens are supported in full screen in native resolutions only (typically 1920x1080). Different resolutions may require separate validation procedures and must be verified in advance with PeriGen personnel.
IMPORTANT: Interaction and acknowledgement with Fetal Alerts must be done on designated workstations where their MAC address is fixed and mapped in the PeriWatch administrative configuration. Alternatively, hostnames can be used as unique identifiers. These stations are designated "Alerting Stations" and must typically be fixed thick clients.

Thin-Client Technologies

Citrix or VMware View deployments of PeriWatch Surveillance client application is now supported for all stations providing station hostname running the thin-client instance is uniquely identified for each session. Using the server names or synthetic host names is not supported.
It is the customer responsibility to map the audio channel to the host workstation such that audible fetal alerts can be heard at the point of care.
Citrix server / VMware view server sizing should be aligned with PeriWatch Client requirements and number of concurrent session requirements (see previous section). Precise server and memory recommendations are a function of overall site load. Recommendations will be finalized during system implementation.
Single Sign-On with server EMRs is supported on both thick and thin client deployments, however, in thin-client mode, the application has to be launched independently before SSO functions are engaged.

Network (For Workstations and Data Acquisition)

Switched Ethernet network, 100 Mbit or better
When used with PeriGen's native DAS/Wall-plate combination or with DIGI/Moxa terminal server, the system requires CAT5+ dedicated cabling between each monitoring area and networking closet. Point-to-point configuration between the monitoring location and the patch panel in the closet is highly recommended.
Proprietary cables between the fetal monitor and wall-plate are provided by PeriGen.

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Tracing Data Acquisition

For the full surveillance system, PeriGen provides a proprietary, secure and resilient solution dedicated to the store and forward of tracing data across the system:

- PeriWatch DAS™, a 16 port RS485 to Ethernet device with internal cache. The total number of DAS devices should be determined by total number of monitoring locations, physical vicinity to the networking closets and a 75% port utilization for port redundancy.
- PeriWatch Single gang wall-plate – For bedside wall mounted data collection from a single monitor (with Ethernet pass-through). Typically, one wall-plate is required per monitoring location.
- PeriWatch Dual gang wall-plate – For bedside wall mounted data collection from up to two fetal monitors (with Ethernet pass-through). Dual gang wall-plates are optional.

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Fetal Monitor Compatibility

Vendor	Protocol compatibility	Type	Interface card / Firmware	Description
Philips	1371	* 1350A and B * 50XM series	M1350-66532 or M1350-66536 * Firmware: **C01.03 and above**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
Philips	1371	Avalon FM20 Avalon FM30 Avalon FM40 Avalon FM50	Firmware: **F.0X and above.**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
DIAP Communication Protocol	RS232	Datascope Passport, Accutor and Spectrum Maternal Monitors	RS232	All compatible maternal vitals sign monitors
GE/Corometrics	1371	120/170/250 Series	J09-J10-J11 Software version 1.3	RJ11 RS232 serial interface

* Philips 1350 A/B and 50XM are presently out of support by Philips. Although the serial protocol is still supported, PeriGen cannot be held responsible for known issues with these fetal monitors with firmware versions lower than **C01.03.** Known issues include a very high frequency of SP02 measurements sent to the electronic tracing when the paper recorder door is open, and no paper is in the-0 tray.

DAS Specifications

Processor Subsystem: Intel® Atom E3815, Single Core

Mainboard: ADLINK Q7-BT1-2G-8G-ER Qseven module

Memory Slot: 2 GB DDR3L

Input / Output:

Two RS-232 ports (DB9 Adapters)

Two USB 2.0 ports

HDMI video interface

16 RS-485 ports with RJ45 connectors and +24VDC power for peripherals

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Externally Accessible CFAST Socket

Storage: 8GB SLC industrial temperature CFast card (includes CE core license)

Network Interface: 1 Gigabit Ethernet Port

Power: 100 – 240 VAC, 60 Hz, 2 A input

Operating Temperature: 0 – 40 Degrees Celsius

Enclosure:

1U, 19” rack mount compatible enclosure

Power LED located on front chassis

Dimensions: 16.75" L x 9.0" W x 1.75" H (19.0" L x 9.0" W x 1.75" H w/ ears)

Warranty: Three years from date of shipment

Regulatory:

UL 60950-1/R:2014-10

CAN/CSA C22.2 No. 60950-1/A2:2014

EN: 55024:2015

EN: 55032:2015

EN: 61000-3-3:2013

FCC, 47 CFR, Part 15 Subpart B

ISED Canada ICES-003 Issue 6

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Network ports used by the system and accessories devices								
	Direction relative to Source device	**Source**	**Destination**	**Protocol**	**App Protocol**	**Destination Port**	**PHI**	**Description**
Core system services	Outbound	DAS, Workstations	App Server	TCP	HTTPS	443	Yes/ Secure	Mandatory. All Communications with the Repository back-end
	Outbound	DAS (Hardware)	App Server	TCP	HTTP	80	No	Mandatory. ONLY for Hardware (rack mounted) DAS. For DAS software updates.
	Outbound	DAS	App Server	TCP	Proprietary	8000	No	Mandatory. Sends data acquired from the Physiological monitors over the TCP connection. Optional for time synchronization to the main app server
	Outbound	DAS (HL7)	App Server	TCP	Proprietary	8010	No	Mandatory. Vitals Data received from EMR data source sent to the main app server.
	Outbound	Workstations	App Server	TCP	Proprietary	8002	No	Mandatory. Receives data acquired from the Physiological monitors over the TCP connection
	Outbound	Workstations	App Server	TCP	Proprietary	8004 (See 8014)	Yes	Mandatory. Connection initiated by source. Events are communicated using this connection.
	Outbound	Workstations	App Server	TCP	Proprietary	8014	Yes / Encrypted	Mandatory. Connection initiated by source. Events are communicated using this connection.
	Outbound	Workstations	App Server	TCP	HTTP	8005	Yes	Only used by the Newborn Module
	Outbound	Workstations	App Server	TCP	Proprietary	8006	Yes / Secured	Only used by the Newborn Module
	Outbound	Workstations	App Server	TCP	Proprietary	8007	Yes / Secured	Only used by the Newborn Module
	Outbound	Workstations	App Server	TCP	HTTPS	8008	Yes / Secured	Only used by the Newborn Module
	Outbound	Workstations	App Server	TCP	Proprietary	8009	No	Only when using/enabling Single-Sign-On (used by Message Receiver)
	Outbound	DAS, Workstations	App Server	UDP	Proprietary	11002	No	Time synchronization to the main app server
Ports used when Patterns/Cues are	Bi-directional	Decision Support Server	App Server	TCP	Proprietary	7101	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Data communication to the decision support modules
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7802	Yes / Encrypted	Only when Cues/Patterns is purchased/installed. Used for tracings interpretation data feed
	Bi-directional	Workstations	Decision Support Server	TCP	HTTPS	7803	Yes /Encrypted	Only when Cues/Patterns is purchased/installed. Plugins

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

								service, Cues algorithm data feed
Ports used when the Vigilance Platform is installed	Bi-directional	Decision Support Server	Decision Support Server	TCP	Net.tcp	8201	Yes / Secured	Only when Vigilance is purchased/installed. HUB DataCollector, patient information & Cues algorithm data feed
	Outbound	Network Redirectors	App Server	TCP	Proprietary	8012	No	ONLY IF DIGI SERIAL SERVERS ARE USED. Sends data acquired from the Physiological monitors over the TCP connection. Static local ports 9000-9XXX used by NRs by default. XXX is number of NRs installed.
	Outbound	Network Redirectors	App Server	TCP	HTTPS	443	No	Used for NR configuration
	Outbound	App Server	Network Redirectors	UDP	Proprietary	9XXX or ephemeral ports (32768-60999)	No	NR and DAS handshake
Ports used when DIGI/MOXA terminal servers are used (typically only with Vigilance standalone system)	Inbound	Moxa Nport 6000 Series	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
	Inbound	Moxa Nport 5000 Series, 6000 Series, Moxa CN2600 Series	App Server	TCP	HTTP	80	No	Device configuration web UI
	Inbound	Moxa Nport 6000 Series, Moxa CN2600 Series	App Server	TCP	HTTPS	443	No	Secure Device configuration web UI
	Inbound	Moxa Nport 5000 Series, 6000 Series, Moxa CN2600 Series	App Server	TCP	Proprietary	950-965	No	Sends data acquired from the Physiological monitors over the TCP connection
	Inbound	Moxa Nport 5000 Series, 6000 Series, Moxa CN2600 Series	App Server	TCP	Proprietary	966-981	No	Command Port
	Inbound	Moxa Nport 5000 Series, 6000 Series	App Server	UDP	Proprietary	4800	No	Broadcast search, Monitoring the status of the NPort, getting the current settings, and Real COM mode mapping
	Inbound	Moxa Nport 5000 Series, Moxa CN2600 Series	App Server	TCP	Proprietary	4900	No	Device Firmware Upgrade Port
	Inbound	Moxa Nport 6610-32	App Server	TCP	Proprietary	950-997	No	Sends data acquired from the Physiological monitors over the TCP connection
	Inbound	Moxa Nport 6610-32	App Server	TCP	Proprietary	966-1013	No	Command Port
	Inbound	Digi	App Server	TCP	Proprietary	771	No	RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection
	Inbound	Digi	App Server	TCP	Proprietary	1027	No	Secure RealPort Service; Sends data acquired from the Physiological monitors over the TCP connection

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

	Inbound	Digi	App Server	TCP	Secure Shell	22	No	Device configuration, secured command line interface
	Inbound	Digi	App Server	TCP	HTTP	80	No	Device configuration web UI

Note: The above list outlines all possible ports with the PeriWatch system (across all possible modules). Always confirm with the PeriGen technical consultant on the applicable ports for your precise configuration.

Anti-Virus Exclusions

All executables and DLL's are digitally signed. This can be used to whitelist the PeriGen application from endpoint protection system.

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.